

TradeDesk AI

Product Vision & UI/UX Master Specification

Complete Design System, User Flows & Screen Wireframes • Target: Field Technicians in India

1. Executive Summary & Value Proposition

The Core Problem

- Manual Paperwork & Lost Revenue: Field technicians finish jobs on-site and write rough paper estimates or forget parts.
- Payment Delays & Uncollected Cash: Technicians lack instant invoicing or follow-up methods, leading to 20-30% delayed bills.
- Complicated Accounting Software: Legacy tools (Tally, Zoho) are built for desktop accountants, not technicians on mobile.

The TradeDesk Solution

1. Multilingual Voice AI: Technician speaks naturally in Hindi, English, or Hinglish to describe work, parts, and rates.
2. Instant GST Invoicing: Automatically calculates labor, materials, and GST taxes into a professional A4 PDF.
3. Dynamic Scannable UPI QR Code: Pre-fills the exact bill amount on customer's Google Pay / PhonePe / Paytm.
4. Instant WhatsApp Delivery: In 1 click, the PDF invoice and payment link are delivered directly to the client's WhatsApp.
5. 1-Click Tax Filing: Exports monthly invoices formatted for GSTR-1 GST accounting.

2. Design System & Visual Foundations

Color Tokens & Semantic Palette

- Background Primary: #0A0F1E (Deep Space Navy - Dark, sleek, glare-free in bright daylight)
- Elevated Card Surface: #111827 / #1F2937 (Frosted surface with #374151 borders)
- Brand Primary Accent: #EA580C / #F97316 (Electric Sunset Orange - High visual prominence)
- Success Green: #22C55E (Paid status, lifetime value, positive growth badges)
- Danger Red: #EF4444 (Unpaid status, delete triggers, mic recording active pulse)
- Warning Amber: #F59E0B (Follow-ups due, overdue payment reminder alerts)
- Typography: Syne/Outfit (Display Headings), DM Sans/Inter (Body/Labels), JetBrains Mono (Financial & Phone data)

3. App Navigation Architecture

Dual Navigation Strategy

- Desktop / Tablet Shell: Collapsible 240px sidebar with Logo, Dashboard, Invoices, Customers CRM, Rate Cards, Settings.
- Mobile Experience (< 768px): Sticky frosted topbar + Bottom navigation bar with central raised Orange Circular Mic FAB.
- Mobile-First PWA: Installable directly to Android and iOS home screens with full offline draft caching capabilities.

4. Screen-by-Screen UI/UX Specifications

Screen 1: Voice-to-Invoice Creation Wizard (/jobs/new)

Step 1: Voice Recording Screen

- Large Central Pulsating Microphone Button: 112px circular touch target with animated ripple rings.
- Live Equalizer Waveform & Timer: Animated jumping audio bars and MM:SS ticker during active recording.
- Real MediaRecorder API: Directly records microphone audio in .webm/.wav format with replay and re-record options.
- AI Multilingual Speech Parser: Extracts client name, 10-digit phone, job title, hours, rate, materials, and GST.

Step 2: AI Review, Signature & Customization

- AI Transcript Preview: Displays raw speech with highlighted structured entity badges.
- Client Information: Customer Name, Phone (+91), Service Address, and B2B Client GSTIN.
- Work & Labor Card: Work title with Quick-Add dropdown to inject items from saved Rate Card catalog.
- Materials & Spare Parts: Line-item editor with real-time price calculations.
- GST Rate Selector: Dropdown for 0% (Nil), 5%, 12%, 18% (Standard), and 28%.
- Digital Signature Canvas: Touch and stylus signature pad for instant customer job acceptance sign-off.
- Real-time Financial Breakdown: Labor + Materials + GST = Total Bill Amount (INR).

Step 3: Success & WhatsApp Dispatch

- Success Card: Green animated verification icon and generated invoice number (e.g. INV-484353).
- 1-Click WhatsApp Delivery: Sends formatted message with direct PDF download link to client mobile.
- Secondary Actions: Direct PDF download and Live Invoice preview link.

Screen 2: Live Digital A4 Invoice & Payment Sheet (/invoices/:id)

A4 Digital Invoice Layout & Payment Engine

- Vendor & Branding Header: Custom business logo, technician name, phone, GSTIN, and business address.
- Client & Invoice Meta: Invoice number, issue date, payment due date, client name, and phone number.
- Itemized Service Breakdown Table: Description, Rate/Unit, Quantity/Hours, Total Amount.
- Dynamic Scannable UPI QR Code: High-resolution QR code generated with upi://pay deep link pre-filling total bill.
- Mobile UPI Intent Button: "Open in UPI App" button to directly launch GPay / PhonePe / Paytm on client phone.
- Verified Digital Signature Block: Displays customer signature stamped during work sign-off.
- Terms & Warranty Footer: Configurable 30-day workmanship warranty and manufacturer parts terms.
- Action Toolbar: Paid/Unpaid toggle badge, Send WhatsApp Reminder, Download PDF.

Screen 3: Customer Mini-CRM & Client Directory (/customers)

Customer Management Features

- Client Directory Search: Instant filtering by client name, 10-digit phone number, or address.
- Dual Column View: Client cards on the left showing total jobs counter and lifetime revenue in rupees.
- Customer Detail Drawer: Detailed service address, custom notes, lifetime spend, and clickable job history timeline.
- Direct WhatsApp Launcher: 1-click button to open WhatsApp chat with any customer.
- Auto-Sync Engine: Creating or updating an invoice automatically syncs/updates customer CRM records.

Screen 4: Rate Cards & Materials Catalog (/catalog)

Rate Cards & Pricing Master

- Category Filtering Tabs: All Items, Services, Labor, Materials & Parts.
- Preset Trade Templates: 1-click auto-loader for AC gas charging, plumbing leaks, MCB switches, and ceiling fans.
- Card Attributes: Item title, category tag, default standard rate (INR), billing unit, and applicable GST rate.
- Invoice Quick Add: Injected directly into New Job line items in 1 click without manual typing.

Screen 5: Technician Command Center Dashboard (/dashboard)

Dashboard Analytics & Actions

- 4 Core KPI Stat Cards: Jobs Today, Month Revenue (Paid INR), Pending Bills (Awaiting payment), Follow-ups Due (>3 days).
- Interactive 7-Day Revenue Trend Chart: Dynamic bars visualizing daily revenue with hover price tooltips.
- 1-Click GSTR-1 CSV Export: Generates formatted CSV file with B2B/B2C, GSTIN, CGST, SGST, and Taxable Values for accountants.
- Overdue Invoice Alert Widgets: Highlights unpaid invoices with 1-click WhatsApp payment reminder buttons.

Screen 6: Business Settings & PDF Customizer (/settings)

Settings & Branding Configuration

- Profile Management: Full name, mobile number, password reset.
- Business Branding: Custom logo image upload (rendered in PDF header), shop name, GSTIN number, invoice prefix.
- UPI VPA Configuration: Setup technician UPI ID (e.g. yourname@okaxis) for dynamic QR code generation.
- Default Warranty & Terms: Configurable footer clauses for all issued invoices.

5. Key Field-Ready UX Principles

Why This UX Wins for Field Contractors

1. Designed for "Dirty Hands": Large buttons, big touch targets, and voice-first inputs minimize typing on site.
2. Zero Accounting Jargon: Plain words (Labor Hours, Rate, Parts Bought, GST) instead of double-entry ledger complexities.
3. Frictionless Customer Payment: Dynamic UPI QR code auto-fills exact amount on GPay/PhonePe to eliminate price haggling.
4. Instant WhatsApp Workflow: No email friction — everything operates natively over WhatsApp.
5. Instant Compliance: GSTR-1 CSV export makes tax filing seamless for local Indian accountants.

